

Theory and Applications of Differential Equations

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Abstract

This article presents a categorical and topological framework for the theory of partial differential equations (PDEs), unifying Green’s functions, Stokes–Ostrogradsky theorems, and Fredholm and Volterra integral equations with applications to swarm coordination. We incorporate the de Rham theorem, simplicial de Rham complexes, and synthetic differential geometry, to formalize local-to-global duality in smooth toposes and differential graded algebras. PDEs are classified (elliptic, parabolic, hyperbolic) and analyzed as morphisms in the category of sheaves, with solutions expressed via integral representations. Swarm coordination is modeled as a dynamic system, with the Cucker–Smale model recommended for its scalability and robustness. This synthesis bridges operational, categorical, and applied perspectives, emphasizing homological structures and their practical implications in robotics, UAV swarms, and biological systems.

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1 Theory of Differential Equations

1.1 Introduction

Differential equations, both ordinary (ODEs) and partial (PDEs), are fundamental to modeling dynamical systems across scientific domains. We approach these equations as structured objects within categorical, topological, and functional-analytic frameworks.

This taxonomy classifies their applications based on their structural properties, topological contexts, and solution methods, drawing on insights from category theory, homology, and synthetic differential geometry.

In the spirit of structural mathematics, partial differential equations (PDEs) are not merely computational tools but objects within a categorical framework, where morphisms reveal universal properties of local-to-global transitions. Green’s functions, Stokes–Ostrogradsky theorems, and Fredholm and Volterra integral equations, together with the de Rham theorem and synthetic differential geometry, form a cohesive structure in the category of functional spaces and smooth toposes. This lecture elucidates these connections, emphasizing homological and operator-theoretic perspectives, and integrates insights from synthetic differential geometry to provide an axiomatic foundation for differential and integral structures.

1.2 PDE as Distributed Dynamical Systems

Definition 1.1. A partial differential equation (PDE) is a relation $L(u) = f$, where $L : \mathbf{E} \rightarrow \mathbf{F}$ is a differential operator between functional spaces $\mathbf{E}$ and $\mathbf{F}$ (e.g., Sobolev spaces $H^k(\Omega)$), and $u, f \in \mathbf{E}$ are sections of a vector bundle over a smooth manifold Ω . In the context of dynamical systems, the parameters of the system (e.g., velocity, temperature) are distributed continuously over Ω , modeled as fields.

PDEs describe systems where dynamics are governed by spatially distributed parameters:

- *Fluid Dynamics:* Navier–Stokes equations model velocity and pressure fields:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \Delta \mathbf{u} + \mathbf{f}.$$

- *Heat Transfer:* The heat equation $\frac{\partial u}{\partial t} = \alpha \Delta u$ models temperature distribution.
- *Quantum Mechanics:* The Schrödinger equation $i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \Delta \psi + V\psi$ describes wave functions.

In the category **Diff** of smooth manifolds, PDEs are morphisms in the category of sheaves of sections of vector bundles. The local-to-global duality is realized through the de Rham complex, where solutions are cohomology classes.

1.3 ODE as Localized Dynamical Systems

Remark 1.1. ODEs model systems where the state variables depend on a single independent variable (typically time), and the dynamics are described by a finite-dimensional state space, not necessarily a discrete set of points. The configuration space may be continuous (e.g., $\mathbb{R}^n$), but the dynamics are localized in the sense that they do not depend on spatial derivatives.

Definition 1.2. An ordinary differential equation (ODE) is a relation $\dot{x} = f(t, x)$, where $x \in \mathbf{M}$ is a state in a manifold $\mathbf{M}$ (e.g., $\mathbb{R}^n$), and $f : \mathbb{R} \times \mathbf{M} \rightarrow T\mathbf{M}$ is a vector field. The dynamics are governed by a single independent variable, typically time.

ODEs model systems with localized dynamics:

- *Mechanical Systems:* The equations of motion, e.g., $\ddot{x} + \omega^2 x = 0$ for a harmonic oscillator.
- *Population Dynamics:* The logistic equation $\dot{x} = rx(1 - x/K)$.
- *Control Systems:* State-space models $\dot{x} = Ax + Bu$.

In the category **Vect** of vector spaces or **Man** of manifolds, ODEs are flows on **M**, represented as morphisms in the category of dynamical systems. The localized nature of ODEs contrasts with the distributed nature of PDEs, reflecting a categorical distinction between finite-dimensional and infinite-dimensional systems.

1.4 Classification of Spaces

Definition 1.3. *Functional spaces for differential equations are classified as follows:*

- Topological Spaces: *Sets with a topology, equipped with continuous maps.*
- Metric Spaces: *Topological spaces with a distance function $d(x, y)$.*
- Linear Spaces: *Vector spaces over $\mathbb{R}$ or $\mathbb{C}$.*
- Normed Spaces: *Linear spaces with a norm $\|\cdot\|$.*
- Banach Spaces: *Complete normed spaces.*
- Euclidean Spaces: *Finite-dimensional Banach spaces with the Euclidean norm.*
- Unitary Spaces: *Hilbert spaces with a unitary group action (e.g., $L_2(\Omega)$).*
- Hilbert Spaces: *Complete inner product spaces (e.g., $L_2(\Omega)$).*
- Fréchet Spaces: *Complete metrizable locally convex topological vector spaces (e.g., $C^\infty(\Omega)$).*

These spaces provide the ambient categories for differential equations:

- PDEs typically operate in Hilbert spaces (L_2) or Sobolev spaces (H^k) due to their completeness and inner product structure.
- ODEs operate in finite-dimensional Euclidean spaces or manifolds.
- Fréchet spaces are used for smooth solutions in $C^\infty(\Omega)$.

1.5 Theory of Distributions

Definition 1.4. *A distribution on a space $\mathbf{E}$ (e.g., $C_c^\infty(\Omega)$) is a continuous linear functional $T : \mathbf{E} \rightarrow \mathbb{R}$ (or $\mathbb{C}$) in the topological dual $\mathbf{E}^*$. The space of distributions is denoted $\mathcal{D}'(\Omega)$.*

- The space ℓ^2 is the Hilbert space of square-summable sequences with the inner product $\langle x, y \rangle = \sum x_i \overline{y_i}$.
- The space $L^2(\Omega)$ is the Hilbert space of square-integrable functions with the inner product $\langle f, g \rangle = \int_\Omega f(x) \overline{g(x)} d\mu(x)$.

Distributions generalize functions, allowing solutions to PDEs with singular sources (e.g., Dirac delta δ_ξ). In $L^2(\Omega)$, solutions to PDEs are often sought due to the space's completeness and orthogonality properties [3].

1.6 Solution Classification of PDE

Definition 1.5. *PDEs of second order are classified into three types based on the eigenvalues of the principal symbol of the differential operator L :*

- *Elliptic: All eigenvalues have the same sign (e.g., Laplace equation $\Delta u = 0$).*
- *Parabolic: One eigenvalue is zero (e.g., heat equation $\frac{\partial u}{\partial t} = \Delta u$).*
- *Hyperbolic: Eigenvalues have mixed signs (e.g., wave equation $\frac{\partial^2 u}{\partial t^2} = \Delta u$).*

Solution methods:

- *Elliptic PDEs:* Solved using Green's functions and Fredholm integral equations in $L^2(\Omega)$. The Green's function $G(x, \xi)$ satisfies:

$$L_x G(x, \xi) = \delta_\xi,$$

yielding solutions:

$$u(x) = \int_{\Omega} G(x, \xi) f(\xi) d\xi + \text{boundary terms}.$$

- *Parabolic PDEs:* Solved using Volterra integral equations or semigroup methods in $H^k(\Omega)$, reflecting causality.
- *Hyperbolic PDEs:* Solved using integral representations or d'Alembert's formula, often in $L^2(\Omega)$.

2 Category Theory of Differential Equations

In the derived category of sheaves, PDE solutions are sections of vector bundles, and the de Rham complex provides a homological framework for integral representations [4].

2.1 Green's Functions: Kernels of Inverse Morphisms

Definition 2.1. *Let $\mathbf{E}$ and $\mathbf{F}$ be Hilbert spaces (e.g., $L_2(\Omega)$ or Sobolev spaces $H^k(\Omega)$), and let $L : \mathbf{E} \rightarrow \mathbf{F}$ be a linear differential operator on a smooth manifold Ω . A Green's function $G : \Omega \times \Omega \rightarrow \mathbb{R}$ (or $\mathbb{C}$) for L is a morphism satisfying:*

$$L_x G(x, \xi) = \delta_\xi,$$

where $\delta_\xi \in \mathbf{F}^*$ is the Dirac delta functional, defined by $\langle \delta_\xi, f \rangle = f(\xi)$ for all $f \in \mathbf{F}$.

The Green's function induces an integral operator $K : \mathbf{F} \rightarrow \mathbf{E}$, given by:

$$(Kf)(x) = \int_{\Omega} G(x, \xi) f(\xi) d\mu(\xi),$$

which acts as a right inverse to L , i.e., $LK = \text{id}_{\mathbf{F}}$ on a suitable subspace of $\mathbf{F}$. In the category of Hilbert spaces, G is a kernel in $\text{Hom}(\mathbf{F}^*, \mathbf{E})$, embodying the duality between $\mathbf{F}$ and its dual $\mathbf{F}^*$.

2.2 Stokes—Ostrogradsky Theorems: Homological Isomorphisms

Theorem 2.1 (Generalized Stokes Theorem). *Let M be an oriented smooth n -manifold with boundary ∂M , and let $\omega \in \Omega^{n-1}(M)$ be a differential $(n-1)$ -form. Then:*

$$\int_M d\omega = \int_{\partial M} \omega,$$

where $d : \Omega^{n-1}(M) \rightarrow \Omega^n(M)$ is the exterior derivative [5].

- *Ostrogradsky–Gauss Theorem:* For a vector field $\mathbf{F}$ on $\Omega \subset \mathbb{R}^n$,

$$\int_{\Omega} \nabla \cdot \mathbf{F} dV = \oint_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} dS.$$

- *Stokes Theorem:* For a vector field $\mathbf{F}$ on a surface $S \subset \mathbb{R}^3$,

$$\int_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{l}.$$

In the category of differential forms $\Omega(M)$, the Stokes theorem is an isomorphism in the de Rham complex, reflecting the exactness of the sequence:

$$\cdots \rightarrow \Omega^{k-1}(M) \xrightarrow{d} \Omega^k(M) \xrightarrow{d} \Omega^{k+1}(M) \rightarrow \cdots.$$

This isomorphism connects local differential structures to global integral structures, enabling the derivation of integral representations for PDE solutions.

2.3 de Rham Theorem and Complex

Definition 2.2 (de Rham Complex). *Let M be a smooth manifold. The de Rham complex is the cochain complex of differential forms:*

$$\Omega^\bullet(M) : \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots,$$

where $\Omega^k(M)$ is the space of smooth k -forms, and d is the exterior derivative, satisfying $d \circ d = 0$. The de Rham cohomology is the cohomology of this complex:

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M))}{\text{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M))}.$$

Theorem 2.2 (de Rham Theorem). *For a smooth manifold M , the de Rham cohomology $H_{\text{dR}}^k(M)$ is isomorphic to the singular cohomology $H^k(M; \mathbb{R})$ with real coefficients:*

$$H_{\text{dR}}^k(M) \cong H^k(M; \mathbb{R}).$$

The isomorphism is induced by the integration map:

$$\omega \mapsto \left(\sigma \mapsto \int_{\sigma} \omega \right),$$

where $\sigma : \Delta^k \rightarrow M$ is a singular k -simplex [4].

In the category of sheaves on M , the de Rham complex is a complex of abelian sheaves:

$$\bar{\mathbf{B}}^k \mathbb{R} = (C^\infty(-, \mathbb{R}) \xrightarrow{d_{\text{dR}}} \Omega^1(-) \xrightarrow{d_{\text{dR}}} \dots \xrightarrow{d_{\text{dR}}} \Omega_{\text{closed}}^k(-)).$$

The de Rham theorem establishes an equivalence in the derived category of sheaves, linking differential forms to topological invariants. In synthetic differential geometry, this complex is internalized in a smooth topos, where the Kock-Lawvere axiom ensures the existence of infinitesimals, facilitating intuitive reasoning about differentials [6].

2.4 Simplicial de Rham Complex

Definition 2.3 (Simplicial de Rham Complex). *Let $X_\bullet : \Delta^{\text{op}} \rightarrow \mathbf{Diff}$ be a simplicial manifold in the category of smooth manifolds $\mathbf{Diff}$. The simplicial de Rham complex is the total complex of the double complex:*

$$\Omega^p(X_q) \xrightarrow{\sum_i (-1)^i \delta_i^*} \Omega^p(X_{q+1}),$$

$$\Omega^p(X_q) \xrightarrow{d_{\text{dR}}} \Omega^{p+1}(X_q),$$

where $\delta_i : X_{q+1} \rightarrow X_q$ are face maps, and d_{dR} is the de Rham differential [7].

The simplicial de Rham complex is quasi-isomorphic to the Chevalley-Eilenberg algebra of the infinitesimal path ∞ -groupoid $\mathbf{\Pi}_{\text{inf}}(X_\bullet)$, reflecting the structure of a geometric ∞ -Lie groupoid. This complex is crucial for equivariant de Rham cohomology and rational homotopy theory, where:

$$C^\infty(\mathbf{\Pi}_{\text{inf}}(X_\bullet)) \simeq \text{Tot } \Omega^\bullet(X_\bullet).$$

2.5 Synthetic Differential Geometry: Axiomatic Framework

Definition 2.4 (Smooth Topos). *A smooth topos $\mathbf{T}$ is a topos equipped with a line object R satisfying the Kock-Lawvere axiom: for any $f : D \rightarrow R$, where*

$D = \{x \in R \mid x^2 = 0\}$ is the infinitesimal object, there exist unique $a, b \in R$ such that:

$$f(x) = a + bx \quad \text{for all } x \in D.$$

A smooth topos models synthetic differential geometry, allowing intuitive reasoning with infinitesimals.

In a smooth topos, the integration axiom posits that the shape modality $\int$ and flat modality $\flat$ define a cohesive structure:

$$\int \dashv \flat \dashv \sharp.$$

The de Rham complex in synthetic differential geometry is internalized as a complex of objects in $\mathbf{T}$, where differential forms are morphisms from infinitesimal thickenings. The integration map in this context is a morphism:

$$\int : \Omega^k(M) \rightarrow \Omega^k(\int M),$$

inducing a synthetic analogue of the de Rham theorem.

2.6 Integral Equations: Operator Algebras

Definition 2.5. Let $\mathbf{E}$ be a Hilbert space, and let $K : \mathbf{E} \rightarrow \mathbf{E}$ be a compact linear operator.

- A Fredholm integral equation of the second kind is:

$$u = f + \lambda Ku,$$

where $f \in \mathbf{E}$, $\lambda \in \mathbb{C}$, and $u \in \mathbf{E}$.

- A Volterra integral equation of the second kind is:

$$u(x) = f(x) + \lambda \int_a^x K(x, \xi) u(\xi) d\xi,$$

where the integral has a variable upper bound.

The kernel $K(x, \xi)$ is often the Green's function $G(x, \xi)$ or its derivatives. For a PDE $Lu = f$ with boundary conditions, the solution may satisfy:

$$u(x) = \int_{\Omega} G(x, \xi) f(\xi) d\xi + \int_{\partial\Omega} K(x, \xi) u(\xi) dS.$$

In synthetic differential geometry, such equations are internalized as morphisms in $\mathbf{T}$, with K defined via the infinitesimal structure of R .

2.7 Structural Unity: Local-Global Duality

The categorical framework unifies these concepts:

- Green’s functions are kernels in $\text{Hom}(\mathbf{F}^*, \mathbf{E})$.
- Stokes–Ostrogradsky theorems are isomorphisms in the de Rham complex.
- The de Rham theorem links differential forms to singular cohomology via integration.
- Simplicial de Rham complexes extend this to ∞ -groupoids.
- Synthetic differential geometry provides an axiomatic framework for infinitesimals and integration.
- Fredholm and Volterra equations are algebraic representations of these morphisms.

This reflects a local-to-global duality, where PDE solutions are cohomology classes in the derived category of sheaves or modules over the ring of differential operators [4, 6].

3 Swarm Coordination Applications

Swarm coordination involves a collection of agents interacting locally to produce emergent global behaviors without centralized control, modeled as a dynamic system using differential equations [?]. This taxonomy classifies swarm models within a categorical framework, leveraging topological spaces, distributions, and homological structures to unify local and global dynamics.

3.1 Swarm Models as Dynamical Systems

Definition 3.1. *A swarm is a collection of autonomous agents in a manifold $\mathbf{M}$ (e.g., $\mathbb{R}^d$) governed by local interaction rules, producing emergent global behavior. The dynamics are modeled as a system of differential equations in the category $\mathbf{Man}$ of smooth manifolds or $\mathbf{Vect}$ of vector spaces.*

3.2 Cucker–Smale Model

Definition 3.2. *The Cucker–Smale model describes N agents with positions $x_i(t) \in \mathbb{R}^d$ and velocities $v_i(t) \in \mathbb{R}^d$:*

$$\dot{x}_i = v_i, \quad \dot{v}_i = \sum_{j=1}^N a_{ij}(x)(v_j - v_i),$$

where $a_{ij}(x) = \frac{K}{(\sigma + \|x_i - x_j\|^2)^\gamma}$ is the interaction weight, and $K, \sigma, \gamma > 0$ are parameters [8].

Remark 3.1. Adaptive control extends the model to handle uncertainties:

$$\dot{v}_i = \sum_{j=1}^N \hat{a}_{ij}(t)(v_j - v_i) + u_i(t),$$

where $\hat{a}_{ij}(t)$ are adaptively updated weights, and $u_i(t)$ compensates for disturbances [9].

Applications include UAV formation control, robotic swarms, and biological flocking [10].

3.3 Vicsek Model

Definition 3.3. The Vicsek model describes self-propelled particles moving at constant speed with alignment:

$$\dot{x}_i = v_i, \quad v_i(t + \Delta t) = |v_i| \frac{\sum_{j \in N_i} v_j}{|\sum_{j \in N_i} v_j|} + \eta_i,$$

where N_i is the set of neighbors within a radius, and η_i is random noise [11].

Suitable for noisy environments but less precise for controlled swarms.

3.4 Alignment-Attraction-Avoidance (AAA) Model

Definition 3.4. The AAA model combines three forces:

$$\dot{x}_i = v_i, \quad \dot{v}_i = \alpha \sum_{j \in N_i} (v_j - v_i) + \beta \sum_{j \in N_i} (x_j - x_i) - \gamma \sum_{j \in N_i} \frac{(x_i - x_j)}{|x_i - x_j|^2},$$

where α, β, γ control alignment, attraction, and avoidance [12].

Used in biological swarms and robotic coordination [15].

3.5 SwarmDMD: Data-Driven Model

Definition 3.5. SwarmDMD uses dynamic mode decomposition to learn inter-agent interactions from observed data, reconstructing swarm dynamics without explicit first-principles modeling [13].

Ideal for reconstructing unknown dynamics but limited for real-time control.

3.6 Hierarchical D-DRL Model

Definition 3.6. The double-layer deep reinforcement learning (D-DRL) model uses inner-layer agents for execution and an outer-layer manager for coordination, modeled as a hybrid system of ODEs and discrete control laws [14].

Suitable for complex tasks but partially centralized.

3.7 Topological and Functional Spaces

Definition 3.7. *Swarm dynamics occur in:*

- Topological Spaces: *Define continuity of agent interactions.*
- Metric Spaces: *Define proximity-based interactions (e.g., $a_{ij}(x)$).*
- Hilbert Spaces: *$L^2(\Omega)$ for mean-field PDE models.*
- Fréchet Spaces: *$C^\infty(\Omega)$ for smooth solutions.*

The communication topology is a graph $G(t) = (V, E(t))$, where vertices V are agents, and edges $E(t)$ are proximity-based connections [15].

3.8 Distributions

Definition 3.8. *A distribution $T \in \mathcal{D}'(\Omega)$ is a continuous linear functional on $C_c^\infty(\Omega)$. For swarm coordination, distributions (e.g., δ_ξ) model singular interactions like local sensing [3].*

Distributions enable modeling of singular forces in PDE-based swarm models (e.g., Dirac delta for collision avoidance).

3.9 PDEs for Large Swarms

Definition 3.9. *For large swarms, the mean-field approximation yields PDEs classified:*

- Elliptic: *Equilibrium configurations (e.g., $\Delta u = 0$).*
- Parabolic: *Diffusion-like spreading (e.g., $\frac{\partial u}{\partial t} = \Delta u$).*
- Hyperbolic: *Wave-like propagation (e.g., $\frac{\partial^2 u}{\partial t^2} = \Delta u$).*

The Vicsek model's mean-field limit is a hyperbolic PDE for density evolution [11].

3.10 Cucker–Smale with Adaptive Control

The Cucker–Smale model with adaptive control is recommended for its balance of simplicity, scalability, and robustness:

- *Scalability:* Handles large N via proximity graphs.
- *Decentralization:* Local rules ensure distributed coordination.
- *Robustness:* Adaptive control handles uncertainties.
- *Applications:* UAV swarms, robotic coordination, biological flocking.

In the category **Man** of smooth manifolds, agents are points, and their dynamics are morphisms defined by the Cucker–Smale equations. The interaction graph $G(t) = (V, E(t))$ induces a simplicial structure, where the simplicial de Rham complex $\Omega^\bullet(X_\bullet)$ models collective behavior as cohomology classes, linking local velocity alignments to global swarm cohesion [7]. In a smooth topos **T**, the infinitesimal interactions (e.g., singular forces modeled by distributions δ_ξ) are internalized via the Kock–Lawvere axiom, providing a synthetic framework for adaptive control laws [6].

3.11 Comparison of Swarm Models

Model	Scalability	Robustness	Computational Complexity
Cucker–Smale	High (proximity graphs)	High (with adaptive control)	Moderate (ODE-based)
Vicsek	Moderate (noise limits)	Moderate (stochastic)	Low (discrete updates)
AAA	High (local rules)	Moderate	Moderate (multiple forces)
SwarmDMD	Low (data-driven)	Low (no control)	High (data processing)
D-DRL	Moderate (hierarchical)	High (learning-based)	High (RL training)

4 Swarm Modeling and Simulation Architecture

The simulation of large-scale drone swarms, as implemented in the Link32 protocol, requires efficient and scalable numerical models to capture the dynamics of 256 to 50,000 drones operating in a shared environment. Two complementary approaches are employed: a discrete ordinary differential equation (ODE) model for individual drone tracking and a continuum partial differential equation (PDE) model for approximating swarm behavior as a density field. These models are implemented in the `skynet_ode.c` and `skynet_pde.c` codes, respectively, using the PETSc library for parallel numerical computations.

4.1 Swarm Modeling Approaches

4.1.1 Discrete ODE Model

The ODE model represents each drone as an individual agent with position $x_i(t)$ and velocity $v_i(t)$ in a 1D or 2D domain, governed by a system of ordinary differential equations:

$$\frac{dx_i}{dt} = v_i, \quad \frac{dv_i}{dt} = u_i + f_{env,i} + f_{swarm,i},$$

where u_i is the control input from the Skynet/Link32 protocol (e.g., velocity adjustments via UDP packets), $f_{env,i}$ is an environmental force (e.g., wind), and $f_{swarm,i}$ accounts for inter-drone interactions, such as repulsion to avoid collisions. The repulsion term $f_{swarm,i}$ is defined as a sum over all other drones $j \neq i$. If the distance between drones i and j , denoted $\|x_i - x_j\|$, is less than a minimum threshold r_{min} , the repulsion force is:

$$f_{swarm,i,j} = \frac{\kappa(r_{min} - \|x_i - x_j\|)}{\|x_i - x_j\| + \epsilon},$$

where $\kappa = 0.05$, $r_{min} = 1.0$, and $\epsilon = 10^{-6}$ ensures numerical stability. Otherwise, if $\|x_i - x_j\| \geq r_{min}$, the repulsion force is zero. The total repulsion force is:

$$f_{swarm,i} = \sum_{j \neq i} f_{swarm,i,j}.$$

This model is suitable for smaller swarms (e.g., 256 drones) where individual tracking is feasible and integrates directly with Skynet/Link32 control signals for precise drone coordination.

4.1.2 Continuum PDE Model

For large swarms (up to 50,000 drones), a continuum approach is adopted, treating the swarm as a density field $\rho(x, y, t)$ with an associated velocity field $u_x(x, y, t)$. The dynamics are governed by a system of partial differential equations, comprising a continuity equation for density conservation and a momentum equation for velocity evolution:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0, \quad \frac{\partial u_x}{\partial t} = f_{control} + f_{env},$$

where $u = (u_x, 0)$ is the velocity field (simplified to 1D flow for initial testing), $f_{control}$ represents Skynet/Link32 control inputs (e.g., density-based velocity adjustments), and f_{env} is an environmental force field (e.g., wind, set to a constant 0.1 in initial simulations). The PDE model is discretized on a 2D grid (e.g., 100×100) using finite differences, with the divergence term approximated as:

$$\nabla \cdot (\rho u_x) \approx \frac{(\rho u_x)_{i,j} - (\rho u_x)_{i-1,j}}{\Delta x}.$$

This approach is computationally efficient for large swarms, capturing collective behavior while integrating Skynet/Link32 control signals to guide the swarm density and velocity fields.

4.2 Architecture of Simulation Codes

Both `skynet_ode.c` and `skynet_pde.c` are implemented using the PETSc library, leveraging its parallel data structures and time-stepping solvers to ensure scalability across multiple MPI processes. The codes are designed to interface with the Skynet/Link32 protocol, which provides control inputs via UDP packets for real-time swarm coordination.

4.2.1 skynet_ode.c Architecture

The `skynet_ode.c` code simulates the discrete ODE model for individual drone dynamics. Its key components are:

- **Data Structure:** A global vector X of size $2N$ (where N is the number of drones, e.g., 17 in the test case) stores positions x_i and velocities v_i . An additional vector `env_data` holds environmental forces (e.g., wind, set to 0.1 per drone).
- **RHS Function:** The `RHSFunction` computes the time derivatives $\frac{dx_i}{dt} = v_i$ and $\frac{dv_i}{dt} = u_i + f_{env,i} + f_{swarm,i}$, incorporating repulsion terms to prevent collisions and Skynet/Link32 control inputs (currently a placeholder, to be updated with UDP-parsed signals).
- **Solver:** The PETSc TS solver with the explicit Runge-Kutta method (TSRK) advances the solution with a time step $\Delta t = 0.1$. The simulation runs for a short duration (e.g., 1 second) to ensure stability during testing.
- **Initialization:** Drones are initialized with random positions in a domain (e.g., $[0, 10]$) using `PetscRandom`, with zero initial velocities.
- **Output:** The final state is written to `skynet_ode.txt`, containing position and velocity pairs for each drone (e.g., 34 values for 17 drones).
- **Parallelism:** The code supports MPI parallelism, distributing drones across processes, though the test output indicates a single process, suggesting a configuration to be optimized for larger swarms.

The code is designed for scalability, with plans to increase N to 256 or 50,000 drones by optimizing MPI distribution and integrating Skynet/Link32 control logic.

4.2.2 skynet_pde.c Architecture

The `skynet_pde.c` code implements the continuum PDE model on a 2D grid. Its architecture includes:

- **Data Structure:** A distributed array (DMDA) manages a 2D grid (e.g., 10×10 in testing, scalable to 100×100) with 2 degrees of freedom per grid point (ρ, u_x). A global vector U stores the state, and a separate vector `env_data` holds environmental forces (e.g., wind, set to 0.1 per grid point).
- **RHS Function:** The `RHSFunction` computes the time derivatives: $\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho u_x)$ using finite differences, and $\frac{\partial u_x}{\partial t} = f_{control} + f_{env}$, with $f_{control}$ as a placeholder for Skynet/Link32 inputs.
- **Solver:** The PETSc TS solver with the explicit Runge-Kutta method (TSRK) advances the solution with $\Delta t = 0.01$. The simulation runs for a short time (e.g., 0.1 seconds) to verify stability.

- **Initialization:** The density ρ is initialized as a Gaussian blob centered at $(50, 50)$ in a 100×100 domain, $\rho(x, y) = e^{-((x-50)^2 + (y-50)^2)/10}$, with zero initial velocity.
- **Output:** The final state is written to `skynet_pde.txt`, containing density and velocity values (e.g., 200 values for a $10 \times 10 \times 2$ grid).
- **Parallelism:** The DMDA structure supports MPI parallelism, distributing the grid across processes, though the test output indicates a single process, requiring optimization for larger grids.

The PDE code is designed to scale to larger grids (e.g., 100×100) and integrate Skynet/Link32 control signals to adjust the velocity field based on swarm density or external inputs.

4.3 Future Enhancements

Both codes are currently stable for small-scale tests (17 drones for ODE, 10×10 grid for PDE). Future work includes:

- Scaling the ODE model to 256–50,000 drones with optimized MPI parallelism.
- Increasing the PDE grid to 100×100 or larger for large swarms.
- Integrating Skynet/Link32 control logic, parsing UDP packets for real-time control inputs (e.g., target positions or velocity fields).
- Developing a hybrid ODE-PDE model using PETSc’s `DMComposite` to combine individual drone tracking with continuum density modeling.

These enhancements will ensure the simulation framework supports the full range of swarm sizes and integrates seamlessly with the Skynet/Link32 protocol for real-time drone swarm coordination.

5 Conclusion

This article synthesizes the theory of partial differential equations (PDEs) and their applications to swarm coordination within a categorical and topological framework. Green’s functions, Stokes–Ostrogradsky theorems, de Rham complexes, and synthetic differential geometry provide a unified perspective on local-to-global duality, with PDE solutions represented as cohomology classes in the derived category of sheaves. Classification of elliptic, parabolic and hyperbolic equations guides solution methods via integral equations, while distributions model singular interactions. Swarm coordination, modeled as a dynamic system, leverages the Cucker–Smale model with adaptive control for its scalability and robustness in applications like UAV swarms, robotic coordination, and biological flocking. This taxonomy bridges operational, categorical, and applied perspectives, offering a rigorous framework for analyzing complex systems [6, 4].

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